1 Number System

Introduction to Digital System

- Digital systems are being used in everyday life, nowadays it is referred as digital age.
- It is a combination of devices designed to manipulate logical information that is represented in digital form.
- Digital systems are easier to design. Information can be easily stored.
- Operations in digital systems can be programmed. They are less affected by noise and can be fabricated on IC chips.
- They are used in communication, traffic control, spacecraft guidance, medical treatment, internet services and many other industrial and commercial applications.
- Nowadays, we have digital mobile phones, digital televisions, digital cameras, and digital computers.
- These devices have graphical user interface (GUI), which makes it easy for the user to interact with them to execute command that appears to the user.
- By using a sequence of instructions (program), operations can be done on the given data. They have the ability to represent and manipulate discrete elements of information.

Introduction to Number System

A set of values used to represent different quantities is called as number system. The number system is a basis for counting various items, for example, a number system can be used to represent the number of students in a class.

Definition

Number system: An ordered set of symbols known as digits is called as number system.

Base of Number System

It is also called as the radix of a number system, denoted by 'r'.

The base/radix of any number system is the total number of digits/symbols used in that number system.

It decides the total number of digits available in that system.

For example, for a decimal number system, the base is 10, i.e., (r = 10).

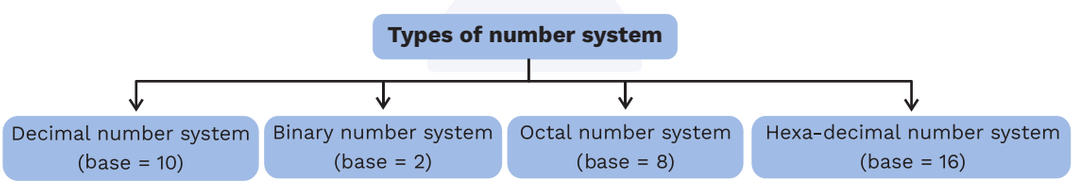


Fig. 1.1

Decimal Number System

- In day to day life, the decimal number system is widely used.
- Base/radix of decimal number system is 10.
- It has 10 unique digits, i.e., from 0 to (r – 1).
- Total digits/symbols = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9).
- For forming the decimal number, the coefficients are multiplied by the powers of 10.

Coefficient	Decimal powers
X_3	10^3
X_2	10^2
X_1	10^1
X_0	10^0
X_{-1}	10^{-1}
X_{-2}	10^{-2}
X_{-3}	10^{-3}

Table 1.1

Definitions

A number system in which the position of each digit of a number has some positional weight is called positional number system.

Example 1:
$X_3 = 3$, $X_2 = 5$, $X_1 = 4$, $X_0 = 2$

Solution:
Decimal number = $(10^3 \cdot X_3) + (10^2 \cdot X_2) + (10^1 \cdot X_1) + (10^0 \cdot X_0)$
Decimal number = $(10^3 \times 3) + (10^2 \times 5) + (10^1 \times 4) + (10^0 \times 2)$
Decimal number = 3000 + 500 + 40 + 2
Decimal number = 3542

Example 2:
$X_2 = 2$, $X_1 = 4$, $X_0 = 6$, $X_{-1} = 5$ and $X_{-2} = 2$

Solution:
Decimal number
$= (2\times10^2) + (4\times10^1) + (6\times10^0) + (5\times10^{-1}) + (2\times10^{-2})$
$= 200 + 40 + 6 + 0.5 + 0.02$
Decimal number = 246.52

Rack Your Brain

If $(290)_{10} = (1205)_r$, then the value of base r is_____.

Binary Number System

- The base/radix of binary system is 2.
- Number of digits used is 2, i.e., '0' and '1'.
- It is a simple system because it consists of only 2 digits, i.e., '0' and '1'.
- Each binary digit is called a bit and group of bits is called word.
- Group of 4 bits is called nibble.
- Group of 8 bits is called byte.
- 1024 bits = 1 K bits.
- So, 64 K bits = 64 × 1024 = 65536 bits.
- If n = number of bits, then total number of combinations = 2^n
- Range of numbers = 0 to $(2^n - 1)$
- The position of 0 or 1 indicates it's weight.
- In a binary number, the weight of each successively higher position bits to the left is an increasing power of 2.

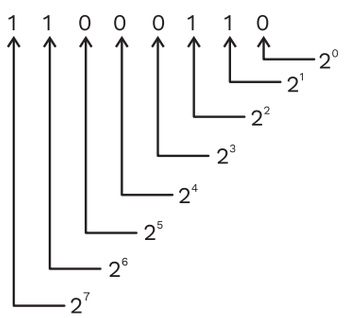
2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
↓	↓	↓	↓	↓	↓	↓	↓
16	8	4	2	1	0.5	0.25	0.125

Definitions

A number system in which the digits of a number does not indicate any significance in position and weight is called non-positional number system.

Example 3:
Convert binary $(11000110)_2$ to decimal $(?)_{10}$.

Solution:
$(11000110)_2 = (1 \times 2^7) + (1 \times 2^6) + (0 \times 2^5) + (0 \times 2^4) + (0 \times 2^3) \times (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0)$
$= 128 + 64 + 0 + 0 + 0 + 4 + 2 + 0$
$= (198)_{10}$



Example 4:
$(1011 \cdot 01)_2 \rightarrow$ Binary

Solution:
$(1011 \cdot 01)_2 = (1\times2^3) + (0\times2^2) + (1\times2^1) + (1\times2^0) + (0\times2^{-1}) + (1 \times 2^{-2})$
$= 8 + 0 + 2 + 1 + 0 + 0.25$
$= (11.25)_{10} \rightarrow$ Decimal

Octal Number System

- The octal number system has base or radix of 8.
- Total number of digits/symbols = 0 to (r – 1), i.e., 0 to 7.
- It is also a positional weighted system.
- Since its base is 8, i.e., 2^3, so every group of 3 binary bits can be represented by an octal digit.
- The octal number system was used in minicomputers.
- Weight of octal system is ascending powers of 8.
- MSB LSB
  8^3 8^2 8^1 8^0

Previous Years' Questions

If given octal number is 66.3, then find the binary number:
(A) 101101.100
(B) 1101111.111
(C) 111111.1111
(D) 110110.011
Solution: (D)

Example 5:
Express octal 1257 to decimal number.

Solution:
Octal number = 1257
$= (1 \times 8^3) + (2 \times 8^2) + (5 \times 8^1) + (7 \times 8^0)$
$= 512 + 128 + 40 + 7$
$= (687)_{10}$
$(1257)_8 = (687)_{10}$
↑ ↑
Octal Decimal

Hexadecimal Number System

- The hexadecimal number system has base of 16.
- Total number of digits/symbols used = 16
- It uses digits:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
↓ ↓ ↓ ↓ ↓ ↓
A B C D E F

- Each bit has a positional weight.
- Weight of LSB = 16^0.
- Weight of higher significant bits is ascending powers of 16.
- Since its base is 16 = 2^4. Hence every 4 digits of binary can be used to represent 1 hexadecimal digit.

Example 6:
Express hexadecimal number $(19FD)_{16}$ into its decimal equivalent.

Solution:
$(19FD)_{16} = 1 \times 16^3 + 9 \times 16^2 + F \times 16^1 \times D \times 16^0$
$(19FD)_{16} = 4096 + 2304 + 240 + 13$
$(19FD)_{16} = (6653)_{10}$
↓ ↓
Hexadecimal number System Decimal number System

Rack Your Brain

Convert hexadecimal number $(1CD.2A)_{16}$ to decimal number.

- Hexadecimal numbers are used for computer due to its short length as compared to binary number system.
- It is used for representing binary so that few errors happen when used to convey information to others.

Conversion of Number Systems

Decimal number system to other base systems

1. Divide the given decimal number by the base of required number system.

2. Get remainder from step 1) as LSB of required base number.
3. Divide the quotient of the step 1 by the new base.
4. Record the remainder from step 3) as next bit of new base number.
5. Repeat steps 3) and 4) until quotient becomes zero.
6. Last remainder will be MSB of new base number.

1. Decimal to binary conversion

Example 7:
Convert decimal number 53 into an equivalent binary number.

Solution:
$(53)_{10} \rightarrow (\ )_2$

2	53		
2	26	1	→ (LSB)
2	13	0	
2	6	1	
2	3	0	
2	1	1	
	0	1	→ (MSB)

So, $(53)_{10} = (110101)_2$

Rack Your Brain

Consider the number by the decimal expression.
$16^3 \times 9 + 16^2 \times 7 + 16 \times 5 + 3$
How many number of 1's present in the above decimal expression?

Example 8:
Convert decimal $(41)_{10}$ to binary.

Solution:

2	41		
2	20	1	(LSB) ↑
2	10	0	
2	5	0	
2	2	1	
2	1	0	
	0	1	(MSB)

$(41)_{10} = (101001)_2$

Rack Your Brain

Given that $(25)_{10} = (100)_b$ what is the value of b?

Note:
- The conversion of a decimal fraction to the binary is obtained by multiplication of 2.
- Keep multiplying the fractional part with 2 until the fraction becomes 0.

Example 9:
Convert $(0.625)_{10}$ to binary.

Solution:

			MSB
0.625	× 2 =		1.25
0.25	× 2 =		0.50
0.50	× 2 =	↓	1.00
			LSB

→ Write integers from top to bottom.
$(0.625)_{10} = (0.101)_2$

Example 10:
Convert $(0.6875)_{10}$ to binary $(\ )_2$.

Solution:

			MSB
0.6875	× 2 =		1.375
0.375	× 2 =		0.75
0.75	× 2 =		1.50
0.50	× 2 =	↓	1.00
			LSB

$(0.6875)_{10} = (0.1011)_2$

Note:
- To convert a decimal fraction to any other base number (base 'r'), similar procedure is used.
- The only change is that instead of 2 multiplication is done by base r.

2. Decimal to octal conversion

Example 11:
Convert $(378.93)_{10}$ to octal $(\ )_8$.

Solution:
Integer part = 378

8	378	
8	47	2 ↑
8	5	7
	0	5

Write the remainders from bottom to top

So, $(378)_{10} = (572)_8$

Fraction part = 0.93

$0.93 \times 8 = 7.44$
$0.44 \times 8 = 3.52$
$0.52 \times 8 = 4.16$
$0.16 \times 8 = 1.28$ ↓

Write the integers from top to bottom.

$(0.93)_{10} = (0.7341)_8$

Therefore $(378.93)_{10} = (572.734)_8$

Rack Your Brain

Given $\sqrt{224} = (13)_r$. What is the value of radix 'r'?

3. Decimal to hexadecimal number

a) To convert given decimal number (integer), divide by 16 till quotient is 0.
b) To convert given decimal fraction, multiply by 16 till we get the required accuracy (number of bits).

Rack Your Brain

Convert $(7864)_{10}$ to its hexadecimal equivalent?

Example 12:
Convert $(2598.675)_{10}$ to hexadecimal.

Solution:

Integer part → 2598

16	2598	Remainder
16	162	6 ↑
16	10	2
16	0	10 → (A)

$(2598)_{10} = (A26)_{16}$

Fraction part → (0.675)

MSB

$0.675 \times 16 = A \leftarrow 10.8$
$0.80 \times 16 = C \leftarrow 12.8$
$0.80 \times 16 = C \leftarrow 12.8$
$0.80 \times 16 = C \leftarrow 12.8$ ↓

LSB

Hence, $(2598.675)_{10} = (A26.ACCC)_{16}$

Example 13:
Convert decimal 725 to hexadecimal.

Solution:

16	725	
16	45	5 → 5
16	2	13 → D
	0	2 → 2 ↓

Therefore, $(725)_{10} = (2D5)_{16}$

Previous Years' Questions

Binary equivalent of hexadecimal number 4F2D is:
(A) 0101111100101100
(B) 0100111100101100
(C) 0100111000101101
(D) 0100111100101101
Solution: (D)

Binary to Other Base Conversion

1. Binary to decimal

Example 14:
$(11011001)_2 \rightarrow (\ )_{10}$

Solution:
Multiply the binary digits with their position weights.

$2^7\ 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0 \leftarrow$ Weight
1 1 0 1 1 0 0 1

$(11011001)_2 = (1 \times 2^7) + (1 \times 2^6) + (0 \times 2^5) + (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$
$= 128 + 64 + 0 + 16 + 8 + 0 + 0 + 1$
$= 217$
Therefore $(11011001)_2 = (217)_{10}$

Example 15:
Convert $(11011.101)_2$ to decimal.

Solution:
$2^4\ 2^3\ 2^2\ 2^1\ 2^0\ 2^{-1}\ 2^{-2}\ 2^{-3} \leftarrow$ weight
1 1 0 1 1 · 1 0 1
$= (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$
$= 16 + 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$
$= 27.625$
Hence, $(11011.101)_2 = (27.625)_{10}$

Example 16:
Convert $(1010.01)_2$ to decimal.

Solution:
$2^3\ 2^2\ 2^1\ 2^0\ 2^{-1}\ 2^{-2} \leftarrow$ weight
1 0 1 0 . 0 1
$= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) + (0 \times 2^{-1}) + (1 \times 2^{-2})$
$= 8 + 0 + 2 + 0 + 0 + 0.25 = 10.25$
$(1010.01)_2 = (10.25)_{10}$

2. Binary to octal

a) An easy way to convert a binary number into octal number is to form a group of 3 binary digits, starting with LSB.
b) Directly write decimal equivalent of those groups.

Example 17:
$(11100101)_2 \rightarrow (\ )_8$.

Solution:

011 100 101
↓ ↓ ↓
3 4 5

Hence, $(11100101)_2 = (345)_8$

Note: Insert bit '0' at MSB side if necessary, to complete a group of 3 bits.

Example 18:
$(1010110101111)_2 = (\ )_8$

Solution:

001 010 110 101 111
↓ ↓ ↓ ↓ ↓
1 2 6 5 7

$(1010110101111)_2 = (12657)_8$

Example 19:
$(10111011)_2 = (\ )_8$

Solution:

010 111 011
↓ ↓ ↓
2 7 3

$(10111011)_2 = (273)_8$

Note: Each octal digit corresponds to 3 binary digits.

Rack Your Brain

Convert $(BC.FBA5)_{16}$ to decimal.

3. Binary to hexadecimal

a) An easy way to obtain hexadecimal number directly from binary number is to make groups of 4 binary digits from LSB.
b) Now, directly write the equivalent decimal of those set of groups.

Note: Each hexadecimal digit corresponds to 4 binary digits.

Example 20:
$(11100101)_2 = (\ )_{16}$

Solution:

1110 0101
↓ ↓
14(E) 5

Hence, $(11100101)_2 = (E5)_{16}$

Numbers with Different Bases

Decimal (r = 10)	Binary (r = 2)	Octal (r = 8)	Hexadecimal (r = 16)
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0010	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Table 1.2 Numbers with Different Bases

Previous Years' Questions

What is the base b of the numbers for the following operation to be correct?

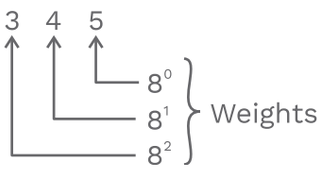
$$\frac{(54)_b}{(4)_b} = (13)_b$$

(A) 2 (B) 4
(C) 8 (D) 16
Solution: (C)

Octal to Decimal

Example 21:
Convert $(345)_8$ octal number to decimal.

Solution:

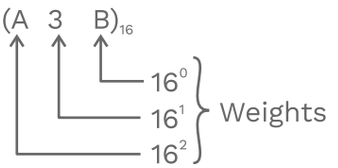


$(348)_8 = (3 \times 8^2) + (4 \times 8^1) + (5 \times 8^0)$
$= 192 + 32 + 5$
$(348)_8 = (229)_{10}$
Octal No. Decimal No.

Hexadecimal to Decimal

Example 22:
Convert $(A\ 3\ B)_{16}$ to decimal.

Solution:
$(A\ 3\ B)_{16}$



$= (A \times 16^2) + (3 \times 16^1) + (B \times 16^0)$
$= (10 \times 256) + (3 \times 16) + (11 \times 1)$
$= (2619)_{10}$
Hence, $(A\ 3\ B)_{16} = (2619)_{10}$

Previous Years' Questions

$(1217)_8$ is equivalent to:
(A) $(1217)_{16}$ (B) $(028F)_{16}$
(C) $(2297)_{10}$ (D) $(0B17)_{16}$
Solution: (B)

Example 23:
Convert $(2\ F\ 3)_{16}$ to decimal.

Solution:

16^2 16^1 16^0 ← Weights
2 F 3

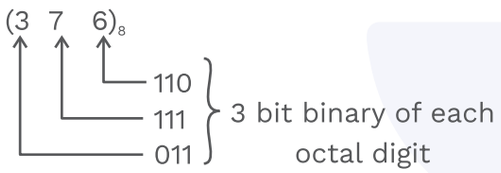
$= (2 \times 16^2) + (F \times 16^1) + (3 \times 16^0)$
$= (2 \times 256) + (15 \times 16) + (3 \times 1)$
$= 512 + 240 + 3$
$= (755)_{10}$
Hence, $(2\ F\ 3)_{16} = (755)_{10}$

Octal to Binary Conversion

- For obtaining the binary equivalent of an octal number, each octal digit is replaced by its 3-bit binary representation.
- Each octal digit represents 3 binary digits.

Example 24:
Convert $(376)_8$ to binary number.

Solution:
Here octal number has 3 unique digits. We just write 3 bit binary equivalent of each octal digits.



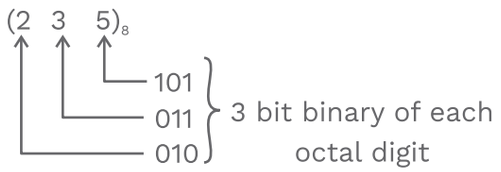
Hence, $(376)_8 = (011111110)_2$

Rack Your Brain

Convert:
$(642.71)_8 = (\ )_2$

Example 25:
Convert $(235)_8$ to binary number.

Solution:



$(235)_8 = (010011101)_2$

Hexadecimal to Binary Conversion

To get binary from hexadecimal number, write 4-bit binary equivalent of individual hexadecimal digits.

Example 26:
Convert $(2\ D\ 5)_{16}$ to binary $(\ )_2$

Solution:



Hence, $(2\ D\ 5)_{16} = (001011010101)_2$

Note: As the base of number system increases, the length of the number decreases.

Miscellaneous Inter-Conversions

Example 27:
Convert $(203)_5$ to decimal $(?)_{10}$.

Solution:
$(203)_5 = (2 \times 5^2) + (0 \times 5^1) + (3 \times 5^0)$
$(203)_5 = 50 + 0 + 3$
$(203)_5 = (53)_{10}$

Rack Your Brain

$(25)_4 + (102)_3 - (21)_5 = (x)_3 + (4)_{x-1}$ where $x > 0$, the value of x is:

Example 28:
Convert $(101011)_2 = (\ )_{10}$

Solution:

Weight → $2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$
↑ ↑ ↑ ↑ ↑ ↑
1 0 1 0 1 1

$= (1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$
$= 32 + 0 + 8 + 0 + 2 + 1$
$= 43$
Hence, $(101011)_2 = (43)_{10}$

Example 29:
Convert $(BAD)_{16} = (\ )_{10}$

Solution:

$$\text{Weight} \rightarrow \underset{\uparrow}{16^2}\ \underset{\uparrow}{16^1}\ \underset{\uparrow}{16^0}$$
$$\text{B}\quad\text{A}\quad\text{D}$$

$= (B \times 16^2) + (A \times 16^1) + (D \times 16^0)$

$= (11 \times 256) + (10 \times 16) + (13 \times 1) = 2989$

$(BAD)_{16} = (2989)_{10}$

Previous Years' Questions

If 73_x (in base–x) is equal to 54_y (in base–y), then the possible values of base-x and base-y are:

(A) 8, 11 (B) 9, 13

(C) 10, 12 (D) 8, 16

Solution: (D)

Example 30:
Convert $(37.4)_8$ to decimal $(\ )_{10}$.

Solution:
Integer part = 37 and fraction part 0.4

$(37)_8 = (3 \times 8^1) + (7 \times 8^0) = 24 + 7 = 31$

and $(0.4)_8 = (4 \times 8^{-1}) = 4 \times 0.125 = 0.5$

Hence, $(37.4)_8 = (31.5)_{10}$

Example 31:
Convert $(3756.42)_8$ to binary $(\ )_2$.

Solution:
Octal → Binary

Write each octal digit into 3-bit binary number.

$(3756.42)_8 = (011111101110.100010)_2$

Example 32:
Convert binary 1101100011.10110 to octal no.

Solution:
Binary → Octal

- Make groups of 3 binary bits, starting from decimal place to left and right directions.
- Insert 0's at last if needed to form groups.

$$\underset{\downarrow}{001}\ \underset{\downarrow}{101}\ \underset{\downarrow}{100}\ \underset{\downarrow}{011}\cdot\underset{\downarrow}{101}\ \underset{\downarrow}{100} = (1543.54)_8$$
$$1\quad 5\quad 4\quad 3\quad 5\quad 4$$

Example 33:
Convert $(565)_{10}$ to hexadecimal $(\ )_{16}$.

Solution:

16 | 565		LSB
16 | 35	5	↑
16 | 2	3	
0	2	MSB

$(565)_{10} = (235)_{16}$

Rack Your Brain

Convert:
$(2B09)_{16} = (\ )_7$

Alternate Method

Convert decimal → binary → hexadecimal.

$(565)_{10}$ → require min 10 bits to represent in binary.

512	256	128	64	32	16	8	4	2	1 ← Weights
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
1	0	0	0	1	1	0	1	0	1

$$(565)_{10} = \underset{\downarrow}{0010}\ \underset{\downarrow}{0011}\ \underset{\downarrow}{0101}$$
$$2\quad 3\quad 5$$

So $(565)_{10} = (235)_{16}$

Rack Your Brain

The solution of quadratic equation $x^2 - 11x + 22 = 0$ are $x = 3$ and $x = 6$. Find the base of the number system?

Example 34:
If $(235)_r = (565)_{10}$, then what is the value of r?

Solution:
$(235)_r = 565$

$(2 \times r^2) + (3 \times r^1) + (5 \times r^0) = 565$

$2r^2 + 3r + 5 = 565$

$2r^2 + 3r - 560 = 0$
After solving equation, we get
$r = 16$

Example 35:
What is the minimum decimal equivalent of 12C?

Solution:
Given number = 12C
Given number 12C has maximum digit C, i.e., base of this number will be 13(D).
So, $(12C)_{13} = (1 \times 13^2) + (2 \times 13^1) + (C \times 13^0)$
$= 169 + 26 + 12 = 207$
$(12C)_{13} = (207)_{10}$

Example 36:
$(4310)_5 = (\ )_{10}$

Solution:
$(4310)_5 = (4 \times 5^3) + (3 \times 5^2) + (1 \times 5^1) + (0 \times 5^0)$
$= 500 + 75 + 5 + 0$
$= 580$
$(4310)_5 = (580)_{10}$

Example 37:
$(198)_{12} = (\ )_{10}$

Solution:
$(198)_{12} = (1 \times 12^2) + (9 \times 12^1) + (8 \times 12^0)$
$(198)_{12} = 144 + 108 + 8$
$(198)_{12} = (260)_{10}$

Binary Addition

1 Bit Addition	
1.	0 + 0 = 0
2.	0 + 1 = 1
3.	1 + 0 = 1
4.	1 + 1 = 0 with carry 1

Table 1.3 Binary Addition

Example 38:
Add the binary numbers 1111 and 1010

Solution:

```
  1111
+ 1010
-------
 11001  ← sum
```

Here
1 + 0 = 1
1 + 1 = 0 with carry 1
1 + 1 + 0 = 0 with carry 1
1 + 1 + 1 = 1 with carry 1
So, (1111) + (1010) = (11001)

Example 39:
Add the binary numbers 1101.1011 and 111.0110

Solution:

```
 +1 +1 +1 +1  +1 +1   ← Carry
   1  1  0  1 .  1  0  1  1
+     1  1  1 .  0  1  1  0
-----------------------------
   1  0  1  0  1 .  0  0  0  1
```

So, (1101.1011) + (111.0110) = 10101.0001

Rack Your Brain

Subtract:
$(1101.1101)_2 - (1001.10)_2$

Binary Subtraction

1 Bit Subtraction	
1.	0 – 0 = 0
2.	0 – 1 = 1, with a borrow of 1
3.	1 – 0 = 1
4.	1 – 1 = 0

Table 1.4 Binary Subtraction

Example 40:
Subtract 1001 from 1101

Solution:

```
  1101
– 1001
-------
  0100
```

Here

$1 - 1 = 0$

$0 - 0 = 0$

$1 - 0 = 1$

$1 - 1 = 0$

Example 41:

Subtract 10 from 1000

Solution:

$$\begin{array}{r} 1000 \\ -\ 10 \\ \hline 0110 \end{array} \rightarrow \text{Result}$$

Here

$0 - 0 = 0$

$0 - 1$ = Cannot be subtracted

So 1 must be borrowed from next non-zero bit, i.e., $10 - 1 = 1$

Example 42:

Subtract $(111.111)_2$ from $(1010.010)_2$

Solution:

$$\begin{array}{r} \rightarrow \text{Borrow} \\ 1\ 0\ 1\ 0\ .\ 0\ 1\ 0 \\ \quad 1\ 1\ 1\ .\ 1\ 1\ 1 \\ \hline 0\ 0\ 1\ 0\ .\ 0\ 1\ 1 \end{array}$$

Previous Years' Questions

Sign extension is the step in:
(A) Signed 16 bit integer addition
(B) Floating point multiplication
(C) Converting a signed integer from one size to another
(D) Arithmetic leftshift
Solution: (D)

Rack Your Brain

Add the following two numbers:
$(7571)_8$ and $(4176)_8$

Octal Addition

Example 43:

Add $(247)_8$ and $(225)_8$

Solution:

$$\begin{array}{r} {\scriptstyle +1 \leftarrow \text{carry}} \\ 2\ 4\ 7 \\ +\ 2\ 2\ 5 \\ \hline 4\ 7\ 4 \end{array}$$

Here

$\rightarrow 7 + 5 = 12$ cannot be in octal, i.e., 12%8 = 4 with carry 1.

$\rightarrow 1 + 4 + 2 = 7$

$\rightarrow 2 + 2 = 4$

Example 44:

Add $(242)_8$ and $(237)_8$

Solution:

$$\begin{array}{r} {\scriptstyle +1+1 \rightarrow \text{carry}} \\ 2\ 4\ 2 \\ +\ 2\ 3\ 7 \\ \hline 5\ 0\ 1 \end{array} \rightarrow \text{sum}$$

Here

$\rightarrow 2 + 7 = 9 \rightarrow$ cannot be in octal, i.e., $9 - 8 = 1$ with carry 1

$\rightarrow 1 + 4 + 3 = 8 \rightarrow$ invalid octal, i.e., $8 - 8 = 0$ with carry 1

$\rightarrow 1 + 2 + 2 = 5$

Previous Years' Questions

11001, 1001 and 111001 correspond to the 2's complement representation of which one of the following set or numbers?
(A) −6, −6 and −6
(B) 25, 9 and 57
(C) −25, −9 and −57
(D) −7, −7 and −7
Solution: (B)

Complement of Numbers

- Complements are used to represent negative numbers.
- It is one of the simplest method to represent negative numbers.
- Complements of any number system can be of two types:
  - Diminished radix complement or (r–1)'s compliment.
  - Radix complement or r's complement.
- **Diminished radix complement**
  It is defined as $(r^n - 1) - N$.
  Where r = Base/radix of system
  n = Number of digits
  N = Given number
- **Radix-complement**
  It is defined as: $(r^n - N)$ for a n-digit number N in base r.

Note: r's complement can be obtained by adding 1 to the (r – 1)'s complement.

Previous Years' Questions

Find the 2's complement of –17.
(A) 110001 (B) 101110
(C) 111110 (D) 101111
Solution: (B)

(r – 1)'s Complement

To determine (r – 1)'s complement, subtract the given number from the highest number in the given base.

Example 45:

1's complement of 10101.

Solution:

5 digit highest number = 11111

So, 1's complement of 10101

$$= \begin{array}{r} 11111 \\ -\ 10101 \\ \hline 01010 \end{array} \leftarrow \text{1's complement}$$

Example 46:

7's complement of 1253.

Solution:

4 digit highest number in octal is 7777

So, 7's complement of 1253 =

$$\begin{array}{r} 7777 \\ -\ 1253 \\ \hline 6524 \end{array} \leftarrow \text{7's complement}$$

Example 47:

9's complement of 5467.

Solution:

4 digit highest number in decimal = 9999

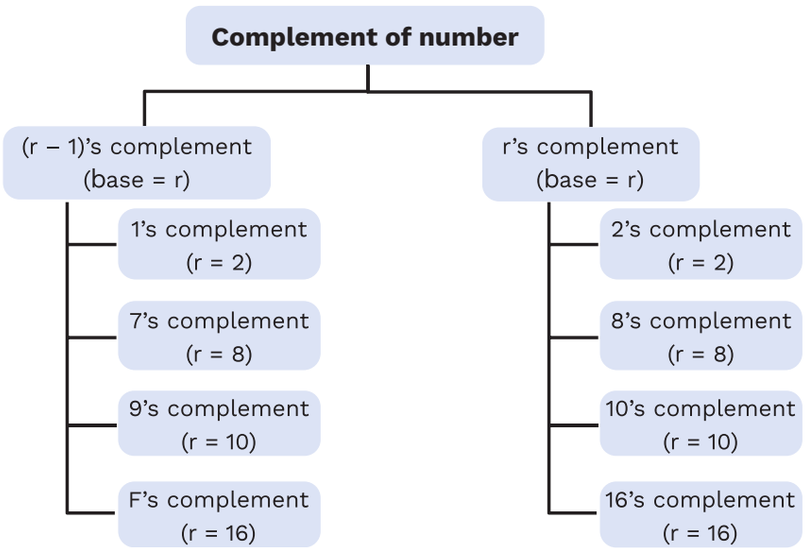


Fig. 1.2

So, 9's complement of 5467 =

$$\begin{array}{r} 9999 \\ -\ 5467 \\ \hline 4532 \end{array} \leftarrow \text{9's complement}$$

Rack Your Brain

Using 10's complement, subtract 3250 from 7352.

Example 48:

F's complement of 3A7D.

Solution:

4 digit highest number in hexadecimal = FFFF

So, F's complement of 3A7D =

$$\begin{array}{r} FF\ FF \\ -\ 3A7D \\ \hline C582 \end{array} \leftarrow \text{F's complement}$$

Note:

- The 1's complement of binary no. can be obtained by changing bit '1' to bit '0' and bit '0' to bit '1'.
- The (r – 1)'s complement of decimal, octal and hexadecimal numbers is obtained by subtracting each digit from 9, 7 and F(15), respectively.

Previous Years' Questions

X = 01110 and Y = 11001 are two 5-bit binary numbers represented in two's complement format. The sum of X and Y represented in two's complement format using 6 bits is:

(A) 000111 (B) 101001
(C) 100111 (D) 001000

Solution: (A)

r's Complement (Radix Complement)

r's complement can be obtained by:

1. First find (r – 1)'s complement
2. Add 1 to the right most digit of (r – 1)'s complement.

Example 49:

2's complement of 10010

Solution:

1's complement of 10010 =

$$\begin{array}{r} 01101 \\ +\ 1 \\ \hline 01110 \end{array} \leftarrow \text{2's complement}$$

Rack Your Brain

Given binary numbers X = 1010101 and Y = 1000010. Perform X–Y and Y–X using 2's complement and 1's complement method.

Shortcut Method (Only for Binary Number)

- **1's Complement** → Invert/change each bit.
- **2's Complement** → Starting from LSB, search for first '1', after that invert all bits.

Example 50:

8's complement of 2671.

Solution:

1. 7's complement of 2671 =

$$\begin{array}{r} 7777 \\ -\ 2671 \\ \hline 5106 \end{array} \leftarrow \text{7's complement}$$

2. Add 1 to right most digit (RMD) =

$$\begin{array}{r} 5106 \\ +\ 1 \\ \hline 5107 \end{array} \leftarrow \text{8's complement}$$

Example 51:

10's complement of 5691.

Solution:

1. 9's complement of 5691

$$\begin{array}{r} 9999 \\ -\ 5691 \\ \hline 4308 \end{array} \leftarrow \text{9's complement}$$

2. Add 1 to RMD

$$\begin{array}{r} 4308 \\ +\ 1 \\ \hline 4309 \end{array} \leftarrow \text{10's complement}$$

Previous Years' Questions

An equivalent 2's complement representation of the 2's complement number 1101 is:

(A) 110100 (b) 001101
(C) 110111 (d) 111101

Solution: (D)

Rack Your Brain

Find the 16's complement of C4AE.

Example 52:

16's complement of 2687.

Solution:

1. F's complement

$$\begin{array}{r} FFFF \\ -2687 \\ \hline D978 \end{array} \leftarrow \text{F's complement}$$

2. Add 1 to RMD =

$$\begin{array}{r} D978 \\ +\ 1 \\ \hline D979 \end{array} \leftarrow \text{16's complement}$$

Previous Years' Questions

The smallest integer that can be represented by an 8-bit number in 2's complement form is:

(A) –127 (B) 0
(C) –256 (D) –128

Solution: (D)

Note: Representation for any positive number is same in among all the data representations.

Unsigned Magnitude

- It is used to represent only positive numbers.
- Range of unsigned number for n-bit number is 0 to $(2^n - 1)$.

Signed Magnitude

- It is used for both positive and negative numbers.
- A sign bit is used.
- An additional bit called sign bit is placed in front of the number.
- For a positive number, sign bit is '0'.
- For a negative number, sign bit is '1'.

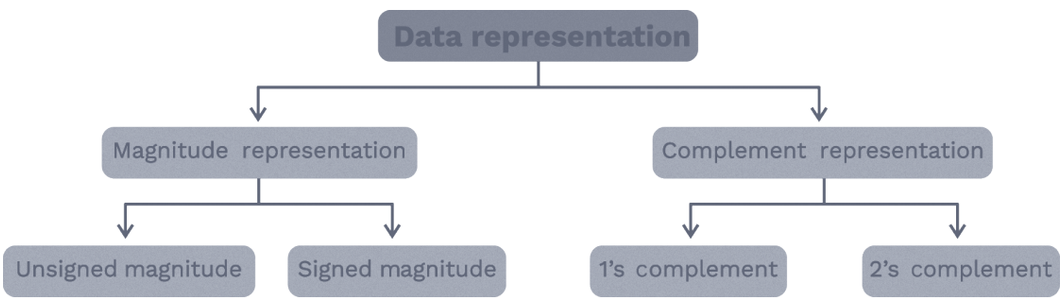
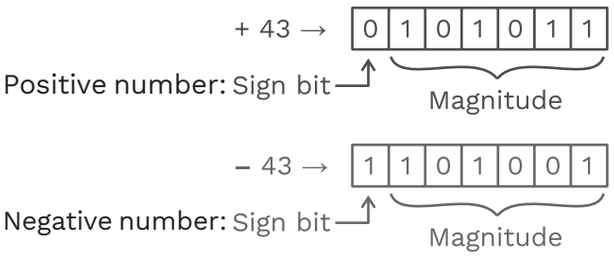


Fig. 1.3

Example 53:

Let a number = 43.



- Range of numbers that can be represented by an n-bit signed number is $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$.

Previous Years' Questions

The range of signed decimal numbers that can be represented by 6-bit 1's complement number is:

(A) Range : –63 to +63
(B) Range : –31 to +31
(C) Range : –32 to +31
(D) Range : –64 to +63

Solution: (B)

1's Complement

1. Using 1's complement representation both positive and negative numbers can be represented.
2. The positive numbers are represented as normal binary with MSB 0.
3. To represent negative number, we have:
   a) Write the positive number in signed magnitude form.
   b) Take 1's complement of complete number (with sign bit).

Example 54:

Express +13 and –13 using 1's complement representation.

Solution:

+13 = 01101

–13 → i) Number in 1's complement = 01101

ii) 1's complement = 10010

So, –13 → 10010

Example 55:

1's complement of a number is 1001. What is decimal equivalent?

Solution:

Given number =

1001 = Negative number
↑
Sign bit = 1, i.e., negative number

1's complement of 1001 = 0110
↑
Magnitude=6

Hence, decimal equivalent = –6

- In 1's and 2's complement representations, to extend the number of bits, sign bit (MSB) can be copied as additional bit.

Previous Years' Questions

A signed integer has been stored in a byte using the 2's complement format. If we want to store the same integer in a 16 bit word. We should:

(A) Copy the original byte to the MSB of the word and fill the LSB with zeros.
(B) Copy the original byte to the LSB of the word and fill the MSB with zeros.
(C) Copy the original byte to the LSB well as the MSB of the word.
(D) Copy the original byte to the LSB of the word and make each bit of the MSB equal to the most significant bit of the original byte.

Solution: (C)

For example:

1's complement of a number is 1001 → 4 bit and 1's complement of another number is 11001 → 5 bit.

But decimal equivalent of both the numbers will be –6.

- In signed magnitude, we insert '0's between sign and magnitude bits to extend the number of bits.

For example: Let number = $\boxed{1}101 = -5$

↑ Sign bit

To increase the number of bits = 10101 = –5 in 5-bit binary.

And in 6-bit representation = 100101 = –5 in 6-bit binary:

- Range of 1's complement = range of signed magnitude representation.
- Range = $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$

Example 56:

Subtract 5 – 3 using 1's complement representation.

Solution:

5 – 3 = 5 + (–3)

Here $\begin{cases} +5 = 0101 \\ +3 = 0011 \\ -3 = 1100 \end{cases}$

So 5 + (–3) = 0101 + 1100 = $\boxed{1}$0001

↓ Carry

Add carry to LSB, we get = 0001 + 1

Hence, 5 + (–3) = → = 0010

Previous Years' Questions

Zero has two representations in:
(A) Only 1 (B) 1 and 2
(C) 1 and 3 (D) 1, 2 and 3
Solution: (B)

Disadvantage of 1's complement and signed magnitude representation:

1. There are 2 representations for '0' (+0 and –0).
   a) In signed magnitude representation: +0 = 0000 & – 0 = 1000
   b) In 1's complement representation: +0 = 0000 & – 0 = 1111
2. During addition, if any carry is present, it must be added to LSB to get the correct result.
3. It makes the circuit complex due to end around carry.

2's Complement Representation

1. There is one unique representation of '0'.
2. The 2's complement of 0 is 0.
3. When a signed number has sign bit as '1' and all magnitude bits are 0's, the decimal equivalent will be -2^n.
   That is, 1000 = decimal equivalent = -2^3 = –8.
4. For a n-bit number the range is: Range – (2^{n-1}) to $(2^{n-1} - 1)$.
5. A negative number may be converted into positive number by taking its 2's complement.

Previous Years' Questions

By using the n bit 2's complement number system, the range of integers that can be represented is:
(A) -2^{n-1} to $(2^{n-1} - 1)$
(B) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
(C) -2^{n-1} to (2^{n-1})
(D) $-(2^{n-1} + 1)$ to $(2^{n-1} - 1)$
Solution: (B)

Example 57:

Express –45 in 8-bit using 2's complement representation.

Solution:

+45 in 8-digit = 00101101

1's complement → = 11010010

+1

2's complement → = 11010011

Here sign bit is 1 that indicates number is negative.

Example 58:

Express –73.50 in 12 bit 2's complement form:

Solution:

+73.50 = 0 1 0 0 1 0 0 1 . 1 0 0 0
1's complement = 1 0 1 1 0 1 1 0 . 0 1 1 1
Add 1 → +1
2's complement → = 1 0 1 1 0 1 1 0 . 1 0 0 0

So, 2's complement of –73.50 is 10110110.1000.

Example 59:

Subtract 15 from 47 using 8-bit 2's complement.

Solution:

We need 47 – 15 = 47 + (–15)

+47 = 00101111 → 8-bit representation

+15 = 00001111 → 8-bit representation

–15 = 11110001 → 2's complement form (–15)

```
  47      00101111
+(–15)  + 11110001
  32    [1]00100000
        carry
```

Carry can be ignored, so result is 00100000

Here, sign bit is 0 hence It is a positive number.

+ 32 = 00100000

Previous Years' Questions

The 2's compliment represen-tation of the decimal value –15 is:

(A) 1111 (B) 11111
(C) 111111 (D) 10001

Solution: (D)

Example 60:

Add –76 to +25 using 8-bit 2's complement.

Solution:

+76 = 01001100

–76 = 10110100 ← 2's complement form (–76)
and +25 = 00011001

So –76 + 25 = 1 0 1 1 0 1 0 0
+ 0 0 0 1 1 0 0 1
No carry →1 1 0 0 1 1 0 1

Since there is no carry in the result, we have to take the 2's complement of the result.

Correct result → 2's complement of 11001101 = 00110011.

Definitions

Alphanumeric codes:
Codes that represent numbers and alphabetic characters (or letters) for conveying information.
ASCII, EBCDIC is alphanumeric codes.

Binary Codes

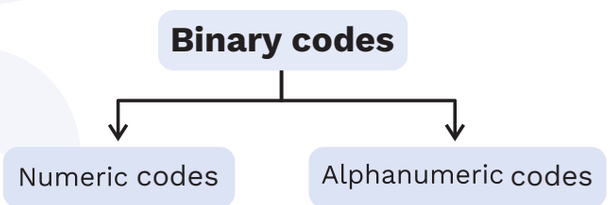


Fig. 1.4

Numeric codes: Numeric information only, i.e., numbers as groups of '0' and 1.

Alphanumeric codes: Alphanumeric information, i.e., letters of alphabet and decimal numbers as set of '0' and 1.

Binary-Coded Decimal (BCD)/8421 Code

- They are used to represent decimal digits.
- In BCD code each digit is encoded into binary.
- BCD combines characteristics of both decimal and binary.
- BCD codes may be weighted or non-weighted.
- Each decimal digit → 4-bit binary equivalent.
- It is also called 8421 code.

Definitions

Weighted codes: In weighted codes, each digit is assigned a specific weight according to its position.
8421 BCD, 2421 BCD and 5211 BCD are weighted codes.

Binary-coded decimal (BCD) of decimal numbers:

Decimal Digit	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Table 1.5

Example 61:

Write the BCD of decimal 186.

Solution:

Decimal number = $\underset{0001}{1}\ \underset{1000}{8}\ \underset{0110}{6}$

So, BCD of 186 = $(000110000110)_{BCD}$

Example 62:

Express (9268) in BCD code.

Solution:

$\underset{1001}{9}\ \underset{0010}{2}\ \underset{0110}{6}\ \underset{1000}{8}$

So, $(9268)_{10} = (1001001001101000)_{BCD}$

Previous Years' Questions

Decimal $(43)_{10}$ in hexadecimal and BCD number system is respectively:
(A) B2, 01000011 (B) 2B, 01000011
(C) 2B, 00110100 (D) B2, 01000100
Solution: (B)

Advantages of BCD

- It is very easy to convert from decimal to BCD.
- It is easy to remember.

Rack Your Brain

Represent decimal number 625 in BCD, Excess-3 and 2421 code.

Disadvantage Of BCD

BCD addition and subtraction is complex.

For 4-bit binary format:

Total possible combination = $2^4 = 16$

Valid BCD codes = 10

Invalid BCD codes = 6

Invalid codes = 1010, 1011, 1100, 1101, 1110, 1111

For 8-bit binary format:

Total possible = $2^8 = 256$

Valid BCD = 100

Invalid BCD = 256 – 100 = 156

BCD Arithmetic

Example 63:

Add 4 to 5 in BCD code.

Solution:

Decimal			BCD
4	→		0100
5	→	+	0101
Decimal 9		→	1001 → Sum in BCD

Grey Matter Alert!

Self-complementing code
Self-complementing codes provide the 9's complement of a decimal number, just by taking 1's complement (invert '1' and '0') in its equivalent code.
2421 code
Excess–3 code

Example 64:

Add 5 and 8 in BCD codes.

Solution:

Decimal		BCD
5	→	0101
8	→ +	1000
13		1101 ← Cannot a BCD digit
	+	0110 add 6 to the sum
carry →	1	0011 (Sum)

Correct BCD sum = 0011

Carry = 1

Example 65:

185 + 575 = (?) in BCD code.

Solution:

Decimal		BCD	
185	→	0001 1000 0101	
575	→ +	0101 0111 0101	
		0110 1111 1010	← BCD sum invalid
		0110 0110	Add 6
		0111 (7) 0110 (6) 0000 (0)	← Valid BCD sum

Definitions

Sequential code: In this code, each succeeding code is one binary number greater than its preceding code.
8421 and Excess-3 are sequential codes.

Example 66:

Find 38 – 16 in BCD code.

Solution:

Decimal		BCD
38	→	0011 1000
16	→	– 0001 0110
BCD differences	→	0010 0010

Excess-3 Code (XS-3) Code

- It is derived from BCD code by adding '3'.
- It is also called 4617 code.
- It is non-weighted code and self-complementing code.
- 1's complement of XS-3 number is the XS-3 code for the 9's complement of that number.

Rack Your Brain

Add the decimal 270 with 491 in BCD, and Excess-3 code.

Excess-3 addition

Following steps are used for Excess-3 addition:

1. Add 3 (0011) in each given number.
2. Add given numbers in Excess-3 form using basic law of binary addition.
3. If there is a carry in the addition, we have to add 3(0011) as well as carry to the sum term to get the final result.

4. If there is no carry in the addition, we have to subtract 3(0011) from the sum term to get the final result.

Example 67:

Add 5 + 8 in XS-3 code.

Solution:

```
Decimal  →  BCD  →   XS–3
1.    5  →  0101 →   0 1 0 1
                   + 0 0 1 1  ← Add 3
                   1 0 0 0  ←XS-3 code
2.    8  →  1000 →   1 0 0 0
                   + 0 0 1 1  ← Add 3
                     1 0 1 1 ← XS-3 code
```

Now add,

```
          1 0 0 0
        + 1 0 1 1
Carry → [1] 0 0 1 1 ← Invalid
          0 0 1 1 ← Add 0011
          0 1 1 0 ← Correct
                    sum in XS-3
```

Example 68:

Add 18 + 39 in XS-3 code.

Solution:

```
18          →        0 0 0 1 1    0 0 0
Add 3       →      + 0 0 1 1      0 0 1 1

XS-3 of 18  →    0 1  0 0 1 0     1 1
39          →        0 0 1 1 1    0 0 1
Add 3       →      + 0 0 1 1      0 0 1 1

XS-3 of 39  →    0 1  1 0 1 1     0 0
      XS-3 of 18 →  0 1  0 0      1 0 1 1
      XS-3 of 39 →+ 0 1  1 0      1 1 0 0

                   1  0 1 0 (1)   0 1 1 1
Propagate carry →    + 1 ↵

                   1  0 1 1       0 1 1 1
Add 3 to correct →   +   0 0 1     1

                   1  0 1 1       1 0 1 0
Subtract 3 to correct
                 →   –   0 0 1 1

XS-3 for 57      →  1  0 0 0 1    0 1 0
```

Excess-3 subtraction

Following steps are used for Excess-3 **subtraction:**

Rack Your Brain

Express the decimal numbers 98, 650 and 813 in the Excess-3 code.

1. Complement the Excess-3 of subtrahend.
2. Add this complemented subtrahend to the minued.
3. If there is a carry, result is positive. In this case we have to add 3(0011) as well as carry.
4. If there is no carry, result is negative. In this case we have to subtract 3(0011).

Definitions

Reflective code: A code is said to be reflective when the code is self-complementing. When the code for 9 is the complement code for 0, the code for 8 is complement for 1, 7 for 2, 6 for 3, and so on.
2421, 5421 and Excess-3 code are the examples of reflective codes.

Example 69:

Subtract 5 from 8 by using XS-3 code.

Solution:

```
5 →                          0 1 01
Add 3 →                    + 0 0 1 1

XS-3 of 5 →                  1 0 00
Complement of 5 in XS-3 →    0 1 11
8 →                          1 0 00
Add 3 →                    + 0 0 1 1

XS-3 of 8 →                  1 0 1 1
XS-3 of 8 →                  1 0 1 1
Complement of 5 in XS-3 →  + 0 1 1 1

                           (1) 0 0 1 0
```

Add 3 → + 0 0 1 1

0 1 01

Add carry → +1

XS-3 of 3 → 0 1 1 0

2421 CODE:

- It is a weighted code.
- It is also a 4-bit code.
- **2421 code for decimal 0–9.**

Rack Your Brain

Form a weighted binary code for the decimal digits (0–9) using weight: 6, 4, 2, 1.

2	4	2	1	Decimal
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
1	0	1	1	5
1	1	0	0	6
1	1	0	1	7
1	1	1	0	8
1	1	1	1	9

Table 1.6

Grey Matter Alert!

Reflected code: It is a sequence of binary numbers, such that two successive bits differ in only one bit. In this code, bits are reflected on both sides of any 2n value. Grey code is a reflected code.

Example 70:

Write 1913 in 2421 code.

Solution:

1 9 1 3
↓ ↓ ↓ ↓
0001 1111 0001 0011

So $(1913)_{10} = (0001\ 1111\ 0001\ 0011)_{2421}$

Three different codes for decimal digits:

Decimal Digit	BCD(8421) Code	2421 Code	Excess-3 Code
0	0000	0000	0011
1	0001	0001	0100
2	0010	0010	0101
3	0011	0011	0110
4	0100	0100	0111
5	0101	1011	1000
6	0110	1100	1001
7	0111	1101	1010
8	1000	1110	1011
9	1001	1111	1100

Table 1.7

Grey Code

- It is a minimum change code/unit distance code.
- It is a reflected code/cyclic code.
- It is non-weighted code.
- It is not suitable with arithmetic operations.
- It is used in input devices.

Definitions

Non-weighted codes: They are not assigned with any weight to each digit position.
Excess-3 and grey codes are non-weighted codes.

Binary to Grey Conversion

Decimal	Binary number ($B_3B_2B_1B_0$)	Grey code ($G_3G_2G_1G_0$)
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Table 1.8

Here 4-bit binary (B_3 B_2 B_1 B_0) is taken and converted to grey code (G_3 G_2 G_1 G_0), we get:

Grey bit **Binary bit**

$G_3 = B_3$
$G_2 = B_3 \oplus B_2$
$G_1 = B_2 \oplus B_1$
$G_0 = B_1 \oplus B_0$

Example 71:

Convert binary $(100101)_2$ in grey code.

Solution:

B_5 B_4 B_3 B_2 B_1 B_0
1 0 0 1 0 1 ← Given binary

For grey code:

1. Copy the MSB.
2. XOR of adjacent pair of binary from left to right.

So, grey code is G_5 G_4 G_3 G_2 G_1 G_0

$G_5 = B_5 = 1$
$G_4 = B_5 \oplus B_4 = 1 \oplus 0 = 1$
$G_3 = B_4 \oplus B_3 = 0 \oplus 0 = 0$
$G_2 = B_3 \oplus B_2 = 0 \oplus 1 = 1$
$G_1 = B_2 \oplus B_1 = 1 \oplus 0 = 1$
$G_0 = B_1 \oplus B_0 = 0 \oplus 1 = 1$
Grey code $\rightarrow$ G_5 G_4 G_3 G_2 G_1 G_0 = 110111

Note: If there is a change in adjacent binary bits, then grey bit is 1.

Grey to binary conversion:

- MSB of binary is same as of grey.
- XOR of each binary with generated grey bit.
- Let 4-bit grey code G_3 G_2 G_1 G_0 and its binary equivalent by B_3 B_2 B_1 B_0, then

Binary bit **Grey bit**

$B_3 = G_3$
$B_2 = B_3 \oplus G_2$
$B_1 = B_2 \oplus G_1$
$B_0 = B_1 \oplus G_0$

Example 72:

Convert the grey code 1011 to binary.

Solution:

G_3 G_2 G_1 G_0 = 1011

So, binary is B_3 B_2 B_1 B_0 = 1101

$B_3 = G_3 = 1$
$B_2 = B_3 \oplus G_2 = 1 \oplus 0 = 1$
$B_1 = B_2 \oplus G_1 = 1 \oplus 1 = 0$
$B_0 = B_1 \oplus G_0 = 0 \oplus 1 = 1$

Chapter Summary



- Number system is required for counting various items.
- The commonly used number systems are decimal, binary, octal and hexadecimal number system.
- These number systems are positional weighted, that means, each digit has a unique weight.
- The base or radix of a number system shows the number of symbols used in that system.
- Bit is the smallest unit of data.
- Combination of 16 bits is called word.
- Complements are used to simplify the subtraction operation.
- Arithmetic operations are more convenient with signed-complement system, for representing the negative numbers.
- Negative numbers can be represented by sign-magnitude, or 1's complement/2's complement form.
- Complement of complement of any number gives original number.
- 1's complement of a binary number can be obtained by inverting each bit.
- 2's complement can be obtained by adding 1 in 1's complement of binary number.
- Binary codes may be numeric or alphanumeric.
- BCD is a numeric code which is used to represent decimal numbers.
- If all weights are positive, binary code is called positively weighted code.
- If at least one of the weights is negative then binary code is called negatively-weighted code.
- In a self-complementing code, binary code for 9's complement of N is the 1's complement of the code for N.
- In a grey code/cyclic code each successive code differs in only one bit position from its preceding code.
- Grey code is called unit distance code.
- XS-3 code can be obtained by adding 0011 in each binary code.

Practice Questions

1. Determine the value of x if $(211)_x = (152)_8$.

(A) 5

(B) 6

(C) 7

(D) 8

2. Find the value of x if $(25)_x = (85)_{10}$.

(A) 10

(B) 20

(C) 30

(D) 40

3. Convert $(4021.25)_{10}$ to binary $(?)_2$.

(A) 111110110100 · 01

(B) 111110110101 · 01

(C) 111110110110 · 10

(D) 111110110111 · 11

4. If a = 101010 and b = 110110, find a – b by 2's complement method.

(A) 110100 (B) 010100

(C) 001100 (D) 100000

5. Convert $(1002{\cdot}403)_8$ to decimal $(\ )_{10}$.
(A) 560 · 558
(B) 562 · 508
(C) 562 · 5058
(D) 562 · 5055

6. Convert $(75.25)_8$ to $(\ )_5$.
(A) $(220 \cdot 130)_5$
(B) $(222 \cdot 131)_5$
(C) $(221 \cdot 130)_5$
(D) $(221 \cdot 131)_5$

7. Subtract 14 from 36 in BCD code.
(A) 00100010
(B) 11000010
(C) 01010000
(D) 00100011

8. Convert binary number 1011011 to grey code.
(A) 100110 (B) 111100
(C) 111011 (D) 110110

9. Convert grey code 11011 to binary number.
(A) 10110
(B) 10010
(C) 10111
(D) 11010

10. Subtract 167.8 from 306.5 in BCD using 9's complement.
(A) 000100111000 · 0111
(B) 010000010100 · 0011
(C) 100001100001 · 0011
(D) 000100111000 · 0011

Practice Solutions

1. Solution: (C) **2. Solution: (D)** **3. Solution: (B)** **4. Solution: (C)**

5. Solution: (C) **6. Solution: (D)** **7. Solution: (A)** **8. Solution: (C)**

9. Solution: (B) **10. Solution: (A)**